# Studio North: customer economics lab

## Are we moving in the right direction as a company?

**Compare acquisition efficiency, contribution value and cash recovery across cohorts.**

A monthly creative-work membership for independent professionals: templates save time, guided learning builds skills, and workshops add value.

24 joining cohorts: January 2024–December 2025.
Monthly renewals; cutoff January 1, 2026 (exclusive).

| | |
|---|---|
| CRM | Renewals |
| Transactions | Product events |
| Feedback | Campaign spend |
| Source credit | Assignments |

# How to read the slide colors

| | |
|---|---|
| **Concept** | Definitions and distinctions to use when interpreting the data. |
| **Case application** | Studio North and its connected customer records. |
| **Evidence interpretation** | Observed results, predictions, and what each can establish. |
| **Group decision** | One focused judgment, discussed with your group. |

# Three ideas guide the entire lab

## 1. Connected customer evidence

The Universal Customer View classifies each dataset's fields as Identifiers, Engagement, Relationship or Perception. It reveals what is recorded and what is missing.

## 2. Customer economics over time

Joining cohorts distinguish customer age from calendar time. Retention, contribution and acquisition cost determine whether relationships create value.

## 3. Incremental marketing value

A randomized campaign comparison asks what changed because of the campaign. A high customer LTV alone cannot answer that question.

**AI proposes the work. Saved SQL, data checks, held-out outcomes and your judgment make the work reviewable.**

# Every dataset gets its own Universal Customer View

Example: people.csv contains one record per person, including prospects who never buy.

| Identifiers | Engagement | Relationship | Perception |
|---|---|---|---|
| person_id<br>age_band<br>household_type<br>region | — | record_created_at<br>attributes_recorded_at | primary_need |

The stated need is a reported intention. Profile timestamps document when the relationship information became available.

**Repeat this field-level table for all eight exports, then open the Archify map of keys, grains and connections.**

# Group decision 1: who needs which onboarding route?

Studio North can offer a ready-to-use template or a guided lesson. Which recorded grouping should inform these onboarding routes?

| Grouping | Two recorded groups | What the field describes |
|---|---|---|
| Age band | 18–34 / 35+ | A demographic characteristic. |
| Stated need | Save time / Build skill | The goal stated before acquisition. |

**Choose one grouping and explain why it is relevant to this onboarding objective.**

**Use the conceptual journey maps to propose the routes. Later, use the SQL results to test whether the grouping explains value.**

# Turn the connected view into a reproducible query

### Inspect raw views

Keep source columns and row meanings.

### Define the customer clock

First paid month sets cohort age index 0.

### Build SQL outputs

Save the exact SQL and ordered results.

### Reconcile the totals

No duplicate money; no missing controls.

## Prompt to try

“Show the SQL that calculates retained customers and contribution by joining month and customer age. Explain the denominator and preserve unobserved future ages as missing.”

**A SQL script makes a calculation inspectable; the data definitions determine whether it answers the right question.**

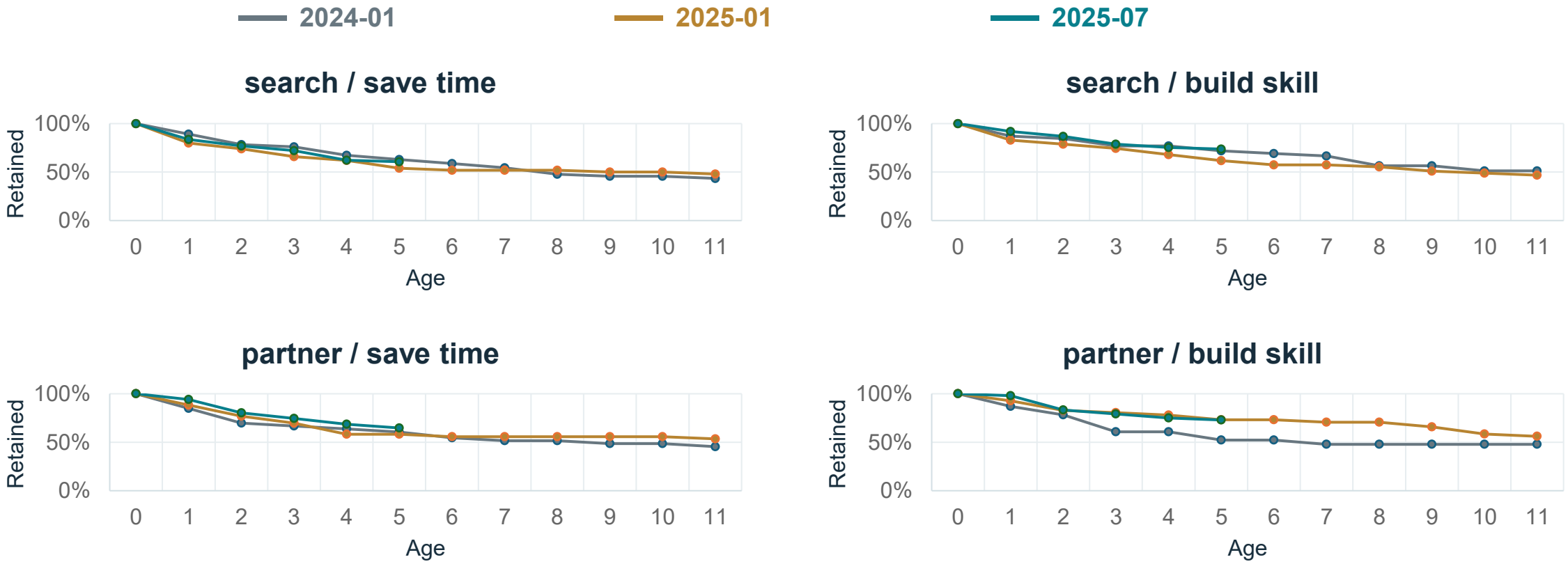
# A cohort follows the same original customers

| Joining cohort | Age 0 | Age 1 | Age 2 | Age 3 | Age 4 | Age 5 |
|---|---|---|---|---|---|---|
| 2024-01 | 100% | 87% | 78% | 72% | 68% | 63% |
| 2024-10 | 100% | 86% | 81% | 75% | 72% | 66% |
| 2025-07 | 100% | 91% | 82% | 76% | 70% | 68% |
| 2025-12 | 100% | — | — | — | — | — |

Across: follow one joining cohort. Down: compare different cohorts at the same customer age.

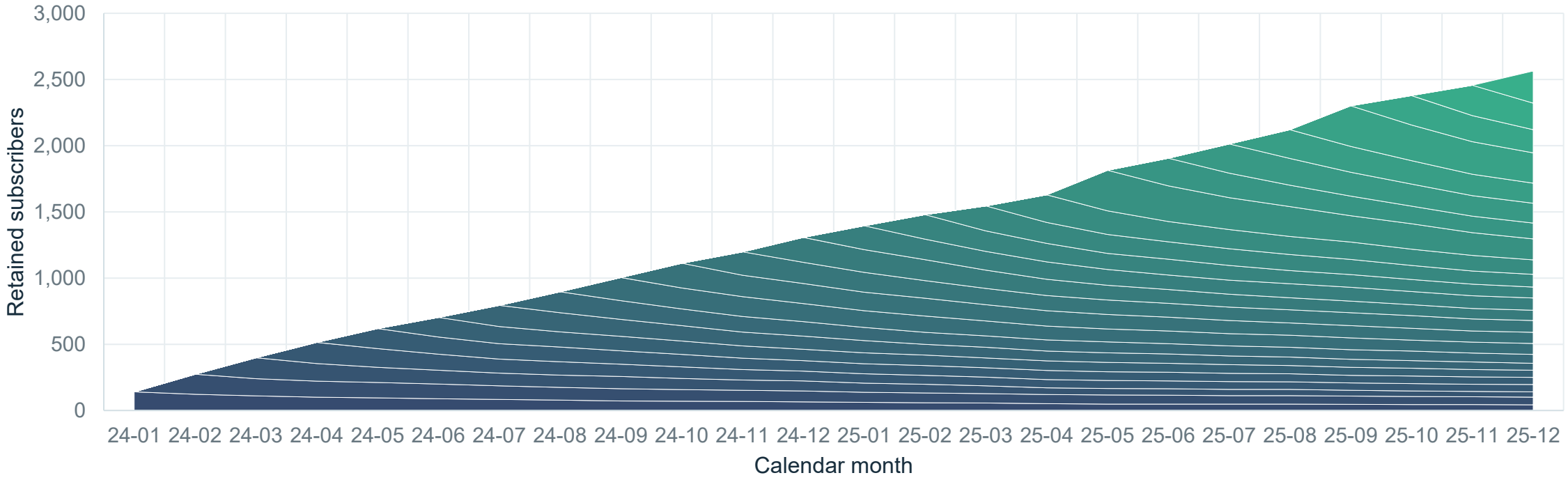
**As of January 1, 2026, December’s cohort has only age 0 observed. Later ages remain blank.**

# Group decision 2: is improvement visible within groups?




**Compare cohorts within each channel and need. Is progress broad, or concentrated in particular groups?**

# Joining cohorts accumulate into the company




**Each band is one joining cohort. The height is the retained customer base, not cumulative acquisitions.**

# Group decision 3: is one month enough to decide?

Each campaign randomized 250 prospects to treatment and 250 to its own control. Follow-up lasts about one month.

| Test | Follow-up | Incremental contribution after media / prospect | Approx. 95% interval |
|---|---|---|---|
| search / save time | 31 days | $1.16 | -$0.62 to $2.94 |
| partner / build skill | 30 days | -$0.36 | -$2.26 to $1.54 |

**Would you scale, stop, or keep measuring? What evidence would change your decision?**

**Cohort payback takes months. Predicted customer LTV cannot replace longer experimental follow-up.**

# A model turns observed renewal histories into an LTV forecast

### Different customer risks

The synthetic business gives each member a persistent baseline churn propensity. A mixture of two distributions creates heterogeneity within segments.

### A simpler fitted model

The main shifted Beta-Geometric model estimates one Beta distribution per cell, with partial pooling across cohorts. It learns from renewals and exits.

### Revenue and value

A Gamma revenue model forecasts spending. Combine retention, contribution and observed months over a common 12-month horizon; subtract CAC for net LTV.

**Separate benchmark: an Orduz-style Bayesian neural network. Compare both models with held-out outcomes and a simple baseline.**

**Describe the model before its results. Generation and inference differ; forecast performance must be tested.**

# Hand your team a decision and a reproducible report

## Are successive cohorts becoming more valuable relative to acquisition cost?

| Report component | What your team should receive |
|---|---|
| Executive summary | The business, a supported verdict, a proposed action and what would change it. |
| Observed and forecast evidence | Same-age retention, revenue, contribution, CAC and cash payback. |
| Model appendix | Model story, parameters, priors, uncertainty and chronological validation. |
| Data and reproducibility appendix | UCV tables, Archify key map and saved SQL; saved manifests accompany the report. |

**The bundled report skill carries the formatting with the project. Each result traces to saved SQL or a documented model.**